

6) COLLECTION OF NIPPLE DISCHARGE AND FEMALE GENITAL TRACT SPECIMENS FOR ROUTINE STAINING

A. GENERAL PRINCIPLES:

- Proper collection is essential for obtaining representative cells and avoiding contamination.
- Explain the procedure to the patient and take consent.
- Use clean, dry, grease-free glass slides.
- Label slides with patient details, site, date and time.
- Prepare smears immediately and fix as per stain used.

IMPORTANT:

- Avoid blood contamination.
- Avoid lubricant or antiseptic before collection.
- Multiple smears increase diagnostic yield.

B. COLLECTION OF NIPPLE DISCHARGE

1. Indications:

- To evaluate nipple discharge for infection, benign lesions (fibroadenoma, duct ectasia) or malignancy.

2. Procedure:

- Clean the nipple and areola gently with normal saline and dry with sterile gauze.
- Do not use spirit, ether or any antiseptic.
- Discard the first drop of discharge.
- Apply gentle pressure around the areola and express the discharge.
- Collect the discharge on clean glass slides.
- Prepare smears immediately.
- Prepare air dried smears for May-Grünwald Giemsa (MGG) stain.
- Prepare alcohol fixed smears for Papanicolaou (PAP) stain.

3. Points to remember:

- Note color, consistency, amount and site of discharge.
- If discharge is scanty, collect after mild stimulation (warm compress) or after galactagogue.
- In bloody discharge, prepare multiple smears.

STAINING OF NIPPLE DISCHARGE SMEARS

Stain	Type of Smear	Fixation	Use
MGG	Air dried	None	Inflammatory, infective conditions
PAP	Wet fixed	95% Ethanol	Epithelial abnormalities, malignancy

POTENTIAL FINDINGS:

- Benign: Ductal epithelial cells, myoepithelial cells, foamy macrophages, neutrophils, histiocytes.
- Infective: Neutrophils, bacteria, fungal elements.

→ Atypia/Malignancy: Atypical ducts cells, clusters, high N:C ratio, prominent nucleoli, pleomorphism.

C. COLLECTION OF FEMALE GENITAL TRACT SPECIMENS

1. Sites from where specimens are collected:

- Exocervix (Ectocervix)
- Endocervix
- Vagina
- Vulva
- Uterus (endometrial sampling)
- Fallopian tube (rarely – by surgical methods)
- Ovary (rarely – by aspiration)

2. Methods of Collection and Procedure:

a) Cervical Smear (Papanicolaou Smear)

- Use Ayre's spatula for ectocervix and endocervical brush or broom for endocervix.
- Insert spatula into the external os, rotate 360° to collect ectocervical cells.
- Insert endocervical brush gently into canal, rotate ¼ to ½ turn and withdraw.
- Immediately smear material on slide.
- Fix immediately in 95% ethanol (for PAP stain).

b) Vaginal Smear

- Use cotton swab or spatula.
- Collect cells from posterior fornix and lateral vaginal walls.
- Smear on slide and fix in 95% ethanol.

c) Vulval Smear

- Use swab or spatula.
- Collect from lesion or surface.
- Smear and fix immediately.

d) Endometrial Smear

- Collected by pipelle, curettage or aspiration.
- Smear material on slide, fix in ethanol.

GENERAL POINTS FOR FEMALE GENITAL TRACT SMEARS:

1. Collect specimen when patient is not menstruating.
2. Avoid sexual intercourse, douching and vaginal medications 48 hours before collection.
3. Use clean instruments and slides.
4. Prepare thin, even smears to avoid overlapping of cells.

**GOOD SPECIMEN COLLECTION + CORRECT SMEAR PREPARATION + PROMPT FIXATION =
ACCURATE CYTOLOGICAL DIAGNOSIS.**

8) CYTOLOGICAL CRITERIA OF MALIGNANCY

Cytological criteria of malignancy are the cellular and nuclear features seen in malignant cells which help in distinguishing them from benign/reactive cells.

A. NUCLEAR FEATURES

1. Increased nuclear size – >3-4 times size of normal intermediate cell nucleus.
2. **High N:C ratio** – Nuclear size large in relation to cytoplasm.
3. **Nuclear hyperchromasia** – Dense, dark staining of nucleus.
4. **Irregular nuclear membrane** – Thickening, notching, infolding.
5. **Coarse chromatin** – Clumping, granular and unevenly distributed.
6. **Prominent/multiple nucleoli** – Large, irregular nucleoli, more than one.
7. **Anisonucleosis** – Variation in nuclear size and shape.
8. **Binucleation/multinucleation**.
9. **Abnormal mitotic figures** – Atypical mitosis, abnormal spindle.

B. CYTOPLASMIC FEATURES

1. Scanty cytoplasm – High N:C ratio.
2. Basophilic cytoplasm – Deep blue staining.
3. Cytoplasmic irregularity – Irregular borders, vacuolations.
4. Keratinization (in some SCC).
5. Diathesis tumor diathesis – Necrosis in background.

★ **No single feature is diagnostic. Diagnosis is based on combination of above features along with clinical correlation.**

DIFFERENCE BETWEEN BENIGN/REACTIVE CELLS AND MALIGNANT CELLS

Feature	Benign / Reactive Cells	Malignant Cells
1. Cell size	Small to intermediate	Large, pleomorphic
2. N:C ratio	Low (1:4 to 1:6)	High (>1:1)
3. Nuclear shape	Round, oval, regular	Irregular, pleomorphic
4. Nuclear membrane	Smooth, uniform	Irregular, thickened
5. Chromatin pattern	Fine, evenly distributed	Coarse, clumped, uneven
6. Nucleoli	Small, inconspicuous	Large, prominent, multiple
7. Cytoplasm	Moderate, well defined	Scanty, irregular, basophilic
8. Mitosis	Normal mitosis absent/rare	Atypical mitosis present

IMPORTANCE:

- Helps in early detection of malignancy.
- Guides further clinical management.
- Used in screening of cancers (e.g. cervix, breast, lung, etc.)

EXAMPLES OF MALIGNANT CELLS:

1. Squamous cell carcinoma (SCC)

2. Adenocarcinoma
3. Small cell carcinoma
4. Undifferentiated carcinoma
5. Melanoma
6. Lymphoma / Leukemia

TYPES OF MALIGNANCY:

1. **Carcinoma** – From epithelial cells.
2. **Sarcoma** – From mesenchymal cells.
3. **Lymphoid malignancies** – e.g. Lymphoma, Leukemia.
4. **Melanoma** – From melanocytes.

Typical Malignant Cell Features:

- Irregular nuclear membrane
- Coarse chromatin
- Prominent nucleoli
- Scanty basophilic cytoplasm

NOTE: Always correlate cytological findings with clinical, radiological and histopathological findings for final diagnosis.

9) CERVICAL CYTOLOGY – NORMAL CERVICAL EPITHELIUM

Cervical cytology is the microscopic examination of exfoliated cells from the uterine cervix to detect normal, inflammatory or neoplastic changes.

A. ANATOMY OF CERVIX:

- Lower part of uterus projecting into the upper vagina.
- Two parts – **Ectocervix** (outside part covered by stratified squamous epithelium) and **Endocervix** (canal lined by columnar epithelium).
- Junction of these two epithelia is called **squamocolumnar junction (SCJ)**.

[Diagram: Uterus → Endocervical canal → Columnar epithelium → Squamocolumnar junction (SCJ) → Squamous epithelium → Vagina. Ectocervix on outside, Endocervix labeled.]

B. NORMAL CERVICAL EPITHELIAL CELLS

Normal cells in cervical smears come from two sources – Ectocervix (squamous cells) and Endocervix (glandular cells).

1. SQUAMOUS EPITHELIAL CELLS (From ectocervix)

These are most frequently seen cells.

Cell type	Size	Nucleus	Cytoplasm
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Superficial squamous cells	Large (50-60 µm)	Small, pyknotic (may be absent)	Abundant, orangeophilic (eosinophilic)
Intermediate squamous cells	Medium (30-50 µm)	Round, small, central	Moderate, cyanophilic (bluish)
Parabasal cells	Small (15-25 µm)	Round to oval, relatively large	Scant, deep blue

FEATURES OF NORMAL SQUAMOUS CELLS:

- Well defined cell membrane.
- Nucleus small with smooth regular outline.
- Fine, evenly distributed chromatin.
- Nuclear : cytoplasmic ratio low.
- Binucleation may be seen in superficial/intermediate cells (normal).
- No hyperchromasia, no abnormal mitosis.

2. GLANDULAR EPITHELIAL CELLS (From endocervix)

Seen as single cells or in groups/sheets.

- Columnar in shape with scant to moderate cytoplasm.
- Nucleus oval to elongated, basal in location.
- Chromatin fine, evenly distributed.
- Nucleoli small or inconspicuous.
- May show cilia or mucin vacuoles.
- Background shows mucus, inflammatory cells (neutrophils, lymphocytes) commonly.

TYPES OF GLANDULAR CELLS:

- Endocervical cells – ciliated, non-ciliated
- Endometrial cells – seen in secretory phase (in groups, syncytial like).
- Metaplastic cells – reserve cells (small, round, scant cytoplasm).

NOTE: In a normal smear, a mixture of above cells is seen with clean background or mild inflammatory exudate.

C. MORPHOLOGICAL FEATURES – NORMAL CERVICAL SMEAR

SQUAMOUS CELLS:

- Superficial – Large, flat, polygonal, eosinophilic cytoplasm, pyknotic nucleus.
- Intermediate – moderate size, cyanophilic cytoplasm, round nucleus.
- Parabasal – small, high N:C ratio, deep blue cytoplasm.
- Well defined borders, no pleomorphism.

GLANDULAR CELLS:

- Columnar cells in single, sheets or honeycomb pattern.
- Nucleus oval, basal, fine chromatin.
- Cytoplasm scant to moderate, may show mucin.
- Ciliated cells may be present.
- Endometrial cells in groups in secretory phase.

BACKGROUND:

- Clean or mucous background.
- May contain neutrophils, lymphocytes, histiocytes in small number.
- RBCs may be present.
- No necrosis or tumor diathesis.
- Absence of abnormal cells.

D. SIGNIFICANCE

- Normal cervical cytology indicates absence of intraepithelial lesion or malignancy.
- Baseline for comparison with abnormal smears.
- Regular screening helps in early detection of precancerous and cancerous lesions.

★ **Normal cervical epithelium shows orderly maturation, minimal nuclear atypia, and absence of dysplastic or malignant features.**

10) MORPHOLOGY OF CELLS IN CERVICAL SMEARS

(Normal, Infections and Neoplastic)

Cervical smear may show normal cells, inflammatory cells, organisms of infection or premalignant or malignant cells. Recognition of their morphology is essential for reporting and diagnosis.

A. NORMAL EPITHELIAL CELLS

1. Superficial squamous cell

- Large, flat, polygonal.
- Abundant pale cytoplasm.
- Nucleus small, pyknotic or absent.
- Borders well defined.
- Stains light green in Pap stain.

[Well defined border, Abundant cytoplasm, Small pyknotic nucleus]

2. Intermediate squamous cell

- Medium size, polygonal.
- Moderate cytoplasm (cyanophilic).
- Nucleus round to oval, central.
- Chromatin fine and evenly distributed.

[Moderate cytoplasm, Central nucleus]

3. Parabasal cell

- Small, round to oval.
- Scant cytoplasm, deep blue (basophilic).
- Nucleus round/oval, relatively large.
- N:C ratio high.

[Scant basophilic cytoplasm, Large nucleus]

NOTE: Maturation sequence of squamous cells from parabasal → intermediate → superficial cells is seen in normal smears.

B. INFLAMMATORY CELLS

- **Neutrophils** – 12-15 μm , multilobed nucleus, fine granules in cytoplasm.
- **Eosinophils** – Bilobed nucleus, orange red granules.
- **Lymphocytes** – Small, round, dense nucleus, scant cytoplasm.
- **Histiocytes (macrophages)** – Large, irregular, abundant cytoplasm, kidney shaped nucleus.
- Background shows mucus, necrotic debris in inflammatory smears.

C. INFECTIOUS ORGANISMS

1. Trichomonas vaginalis

- Pear shaped, 10-20 μm .
- Eccentric nucleus, undulating membrane, axostyle.
- Seen in wet smear.

2. Candida species

- Oval budding yeast cells.
- Pseudohyphae and true hyphae may be seen.

3. Actinomyces (Sulfur granules)

- Basophilic clumps with radiating clubs.

4. Bacterial vaginosis (Clue cells)

- Squamous cells with fuzzy borders due to coating of bacteria (Gardnerella).

D. ABNORMAL / NEOPLASTIC CELLS

1. Atypical Squamous Cells (ASC)

- Enlarged cell size.
- Increased N:C ratio.
- Nuclear enlargement, irregularity.
- Hyperchromasia, coarse chromatin.
- Irregular nuclear membrane.
- May be due to HPV infection, dysplasia.

2. Dysplastic Cells

- Pleomorphism – variation in size and shape.
- High N:C ratio.
- Hyperchromatic nuclei.
- Irregular chromatin distribution.
- May occur in mild, moderate or severe dysplasia.

3. Carcinoma in situ

- Marked pleomorphism.
- Very high N:C ratio.
- Coarse chromatin, irregular membrane.
- Disordered maturation.
- Cells seen singly or in sheets.

4. Invasive Carcinoma Cells

- Markedly pleomorphic.
- Very large, hyperchromatic nuclei.
- Prominent nucleoli.
- Tumor diathesis in background (necrosis, blood, inflammatory cells).

KEY POINTS:

- ★ Evaluate – cell type, size, shape, N:C ratio, chromatin, nucleoli, cytoplasm and background.
- ★ Correlate cytological findings with clinical and histopathological findings.
- ★ Early detection of precancerous and cancerous lesions helps in prevention of cervical cancer.

PAP STAIN – COLOUR OF CELLS:

Cell type	Stain colour
Superficial squamous cells	Light green
Intermediate squamous cells	Cyan / Blue green
Parabasal cells	Deep blue
Nuclei	Dark blue / Purple
Keratin, RBCs	Orange / Pink

N:C RATIO GUIDE:

- ★ Normal cells – Low
- ★ Dysplastic cells – Increased
- ★ Malignant cells – Very high

Thorough knowledge of cell morphology in cervical smears is the cornerstone of accurate cytological diagnosis. ♥

PCR (POLYMERASE CHAIN REACTION)

PCR is a molecular technique used to amplify a specific segment of DNA *in vitro* to millions-billions of copies. It was developed by Kary B. Mullis in 1983.

PRINCIPLE:

- PCR is based on specific annealing of two oligonucleotide primers to sequences flanking the target DNA.
- A thermostable DNA polymerase synthesizes new DNA strands from primers.
- Repeated cycles of denaturation, annealing and extension lead to exponential amplification.

COMPONENTS:

- ① Template DNA – target sequence
- ② Two primers – Forward & Reverse
- ③ Thermostable DNA polymerase – Taq polymerase (from *Thermus aquaticus*)
- ④ dNTPs – dATP, dTTP, dGTP, dCTP
- ⑤ MgCl₂ – cofactor
- ⑥ Reaction buffer – provides optimal pH & ions
- ⑦ Nuclease free water

STEPS OF PCR (THERMAL CYCLING):

1. Denaturation: 94-98°C for 15-30 sec

→ Double stranded DNA separates into single strands.

2. Annealing: 50-65°C for 20-40 sec

→ Primers anneal to their complementary sequences on single stranded DNA.

3. Extension (Elongation): 72°C for 30 sec – 1 min/Kb

→ Taq polymerase extends primers and synthesizes new DNA strand in 5' → 3' direction.

One Cycle:

94-98°C: ds DNA → Denaturation → single strands

50-65°C: 5'...3' + primers → Annealing

72°C: Extension → new complementary strands synthesized

Repeat 25-35 cycles

After 'n' cycles → 2ⁿ copies

30 cycles → ~10⁹ copies

TYPES OF PCR:

- **Conventional (End point) PCR** – detects product after amplification.
- **Real time (qPCR)** – detects and quantifies product during amplification using fluorescence.
- **Multiplex PCR** – amplification of multiple targets in a single reaction.
- **Nested PCR** – two successive PCRs to increase sensitivity & specificity.
- **RT-PCR** – RNA is first converted to cDNA by reverse transcriptase, then amplified.
- **Hot start PCR** – polymerase activated by heat to prevent non-specific amplification.

APPLICATIONS OF PCR:

- Detection of infectious agents (viruses, bacteria, parasites).
- Diagnosis of genetic disorders & mutations.
- Forensic analysis – DNA fingerprinting.
- Paternity / maternity testing.
- Oncology – detection of gene mutations, translocations.
- Gene cloning & expression studies.

- Detection of GMO in food.
- Quantification of gene expression (qPCR).
- Environmental monitoring & microbial identification.

ADVANTAGES:

- Highly sensitive, specific & rapid.
- Requires very small amount of DNA.
- In vitro technique – no living organism needed.
- Can be automated.

LIMITATIONS:

- Contamination can give false positive results.
- Requires expensive equipment & reagents.
- Requires prior knowledge of target sequence.
- Inhibitors in sample may affect results.

★ PCR is a powerful tool in Molecular Biology and Clinical Diagnostics.

4) ELECTROPHORESIS – Principle, Technique & Applications

DEFINITION:

Electrophoresis is the migration of charged particles in an electric field towards the oppositely charged electrode.

TYPES (BASED ON SUPPORTING MEDIUM):

1. Paper electrophoresis
2. Agar / Agarose gel electrophoresis
3. Polyacrylamide gel electrophoresis (PAGE)
4. Capillary electrophoresis
5. Isoelectric focusing (IEF)

TECHNIQUE (GENERAL STEPS):

1. Preparation of supporting medium (paper/gel) & buffer.
2. Sample preparation & application at origin.
3. Placement of medium in the electrophoresis apparatus.
4. Application of voltage (DC).
5. Migration of charged molecules.
6. Staining / detection.

7. Rinsing / destaining (if required).
8. Interpretation and documentation.

PRINCIPLE:

- At a given pH, molecules acquire a net charge.
- When an electric field is applied, they move toward the electrode of opposite charge.
- Rate of migration depends on:
 1. Net charge (q)
 2. Size & shape of molecule
 3. Buffer pH and ionic strength
 4. Nature of supporting medium
 5. Voltage applied

Migration ∝ Net charge / (Size & frictional resistance)

APPARATUS – BASIC SETUP (TANK METHOD):

[Diagram: Cathode (–) on left, Anode (+) on right. Sample applied at origin. Fast moving = more charge/smaller size. Slow moving = less charge/larger size. Buffer at bottom.]

IMPORTANT TERMS:

- Anode (+) : positive electrode
- Cathode (–) : negative electrode
- Origin : point of sample application
- Migration : movement of molecules
- Mobility : distance moved per unit time per unit applied field.

SERUM PROTEIN ELECTROPHORESIS (SPEP):

Principle: Separation of serum proteins on the basis of charge and size at alkaline pH (8.6).

Procedure (Brief):

1. Serum is applied on agarose strip.
2. Electrophoresis is performed.
3. Proteins separate into fractions.
4. Stained with amido black.
5. Densitometry is done to obtain electrophoretogram.

Normal Electrophoretogram:

Alb - Albumin, α₁ - Alpha 1 globulin, α₂ - Alpha 2 globulin, β - Beta globulin, γ - Gamma globulin

PATTERNS (SPEP):

Condition	Characteristic Finding
Normal	Albumin ↑, α ₁ , α ₂ , β moderate, γ broad.
Acute inflammation	α ₁ & α ₂ ↑
Chronic inflammation	γ ↑ (polyclonal)

Liver cirrhosis	β - γ bridging
Nephrotic syndrome	Albumin ↓, α_2 ↑, β ↑, γ ↓
Multiple myeloma	M-spike in γ region
Hypogammaglobulinemia	γ region ↓
α_1 -antitrypsin deficiency	α_1 region ↓

HEMOGLOBIN ELECTROPHORESIS:

Principle: Separation of Hb variants based on difference in charge at alkaline pH (8.6-8.8).

Used for: Detection & identification of abnormal hemoglobins & thalassemia.

Normal Adult Pattern:

Hemoglobin	%
Hb A	96-98 %
Hb A2	2-3.5 %
Hb F	< 1 %

Abnormal Patterns:

- Hb S (Sickle cell anemia)
- Hb C disease
- Hb E trait / disease
- β / α Thalassemia
- Hb D, Hb Lepore etc.

ISOELECTRIC FOCUSING (IEF):

Principle: Separation of proteins based on their isoelectric point (pI) in a pH gradient.

- Proteins migrate in a pH gradient until they reach pI where they carry no net charge and stop.

Applications:

1. Separation of isoenzymes
2. CSF protein analysis
3. Detection of monoclonal proteins
4. Research studies

USES / APPLICATIONS:

1. Clinical diagnosis (SPEP, Hb electrophoresis)
2. Protein & enzyme analysis
3. Detection of genetic disorders
4. Forensic science (DNA profiling)
5. Research in biochemistry & molecular biology

ADVANTAGES:

- Simple & reproducible
- High resolving power

- Small sample required
- Useful for qualitative & quantitative analysis
- Wide range of applications

LIMITATIONS:

- Time consuming (some types)
- Requires skilled personnel
- Some methods have low sensitivity
- Interpretation may be difficult in complex samples

KEY POINTS (MUST REMEMBER):

- ★ Migration $\propto$ Net charge / (Size & frictional resistance)
- ★ At pH > pI $\rightarrow$ protein moves towards cathode (-)
- ★ At pH < pI $\rightarrow$ protein moves towards anode (+)
- ★ SPEP is important for detection of M-protein (monoclonal gammopathy).
- ★ Hb electrophoresis is gold standard for Hb variants.

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